**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 145 · Lesson 145 · Packet Switching, Circuit Switching and Reliability

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the job of DNS?

Q2. What is the accumulator for?

Q3. Which is faster in the worst case on sorted arrays, linear or binary search?

Q4. Which logic gate outputs 1 only when both inputs are 1?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

packet switching — Data divided into packets that may take different network routes.

circuit switching — A dedicated end-to-end circuit established before communication.

latency — Delay experienced by data in transit.

retransmission — Sending data again when loss is detected by a reliable protocol.

### New learning

Packetise data — Large message split into addressed packets for shared links.

Route over network — Packets share capacity and may arrive out of order.

Reassemble reliably — TCP can reorder and retransmit data, while IP itself is best-effort.

### Worked example

A 3-packet file crosses a packet-switched network. Packets arrive 2,1,3. Explain handling.

1. Send packets 1,2,3 — Identify the relevant information and make a prediction.

2. Receive 2,1,3 — Explain the change using correct technical terms.

3. Reorder 1,2,3 — Packets can follow different routes or encounter different queue delays. With a reliable transport such as TCP, sequence numbers permit correct reordering before delivery of the byte stream; missing data can be retransmitted.

## We Do · Practise with guidance

A telephone system reserves capacity for a call before any voice data flows. Compare to ordinary IP networking.

Reserve dedicated circuit

Identify the relevant information and make a prediction.

Hold resources during call

Explain the change using correct technical terms.

Compare shared packet links

A circuit-switched system sets aside end-to-end resources while active; ordinary packet-switched IP shares links dynamically with others. Circuit switching offers predictable reserved capacity but can waste it during silence; packet switching adapts.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Define packet switching and circuit switching, highlighting two differences. [6]

### Question 2

Explain why packets in a message may reach the destination out of order. [6]

### Question 3

Evaluate packet switching versus circuit switching for video calls, including reliability and resource efficiency. [8]

## Exit Ticket · Answer before checking

IP alone guarantees that every packet is received in the original order.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Map names such as example.org to relevant resource records including IP addresses.

Q2. Holding a CPU's working data/ALU results.

Q3. Binary search: O(log n) rather than O(n).

Q4. AND.

### You Do pages · Independent questions

Q1. Packets individually forwarded/shared links [3], circuit pre-establishes reserved route/capacity [3].

Q2. Different routes [2], variable congestion/queuing [2], sequence numbers and reliable transport reorder [2].

Q3. Shared capacity efficiency [2], delay/jitter concerns [2], circuit reservation benefits/costs [2], qualified judgement incl transport/QoS [2].

### Exit Ticket · Model answer

False. IP offers best-effort delivery; transport protocols may provide ordering and retransmission.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**



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BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check understanding